

Team Sudophiles

National Health Resource & Supply Chain Resilience Platform

SmartCare & AushadhiNet Mesh

A federated AI public health ecosystem providing real-time visibility into medicine stocks, bed availability, and medical personnel attendance across India's entire primary health centre network.

420+

PHCS CONNECTED

15 Mins

TELEMETRY SYNC

5 States

FEDERATED AI HUBS

Zero-PII

PRIVACY GUARANTEE

Working Prototype

smartcare-omega-two.vercel.app

Demo Video (Drive)

Public View Access

01 • CHALLENGE ASSESSMENT

The Problem: Fragmentation & Stock-Outs in Public Health

Public healthcare systems across India face persistent supply chain vulnerabilities, delayed paper telemetry, and inability to dynamically respond during health emergencies.

Systemic Healthcare Bottlenecks

Opaque Rural Inventories
Clinics experience unrecorded stock-outs of life-saving medicines (antipyretics, snake antivenom, antibiotics) leading to patient turnaround.

Unpredicted Bed & Oxygen Telemetry
No real-time status of ICU, Oxygen, and General Ward bed occupancy during sudden regional disease outbreaks.

Clinical Attendance Asymmetry
Doctor and nurse shift attendance untracked, causing dangerous Patient:Doctor ratio spikes (>45:1) without administrative awareness.

The SmartCare Solution

Automated 15-Minute Telemetry Sync
Continuous edge telemetry automatically updates medicine stock levels, bed turnover, and medical personnel shifts on state dashboards.

AI Epidemic Surge Forecaster
Simulates seasonal outbreak spikes (Monsoon Dengue, Gastroenteritis, Viral Flu) to predict depletion dates 72 hours in advance.

Multi-State Federated AI (FedAvg)
Aggregates model weights across 5 states without moving citizen health records, guaranteeing zero patient PII exposure under the DPDP Act.

02 • SYSTEM TOPOLOGY

4-Tier Resilient Healthcare Architecture

Seamless decentralized telemetry flow from rural health sub-centres up to the National MoHFW Command Desk.

// TOPOLOGY HIERARCHY

Tier 4: National & State AI Command
MoHFW Federated Hub • 5 State AI Nodes (FedAvg)

VERTEX AI

▲ 15-Minute Gradient Exchange (ΔW) ▲

Tier 3: District CMO & Redistribution Depot
Central Medical Stores • Automated Fleet Routing • OTP Manifest

REBALANCE

▲ Real-Time Telemetry Packets ▲

Tier 2: Primary Health Centres (PHC)
Stock Buffers • Bed Telemetry • Staff Shifts • e-Rx Studio

420 PHCS

▲ Citizen Walk-Ins & 108 SOS ▲

Tier 1: Citizens & Patients
ABHA Health Passport • QR OPD Triage • 108 Emergency SOS

ABDM

Technical Foundations

Google Firebase & Firestore Real-Time Mesh
Multi-tenant synchronization across 420+ facilities with instant offline queueing and background reconciliation.

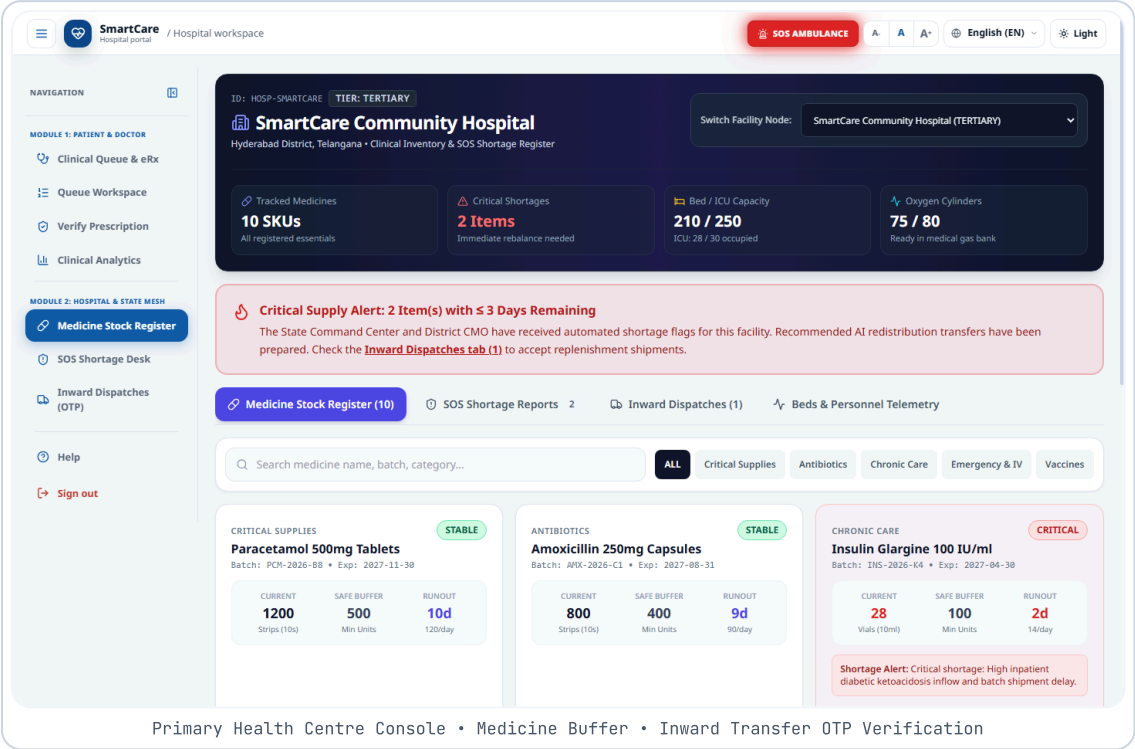
Deterministic Edge ML Execution
Calculates Days-of-Stock and surge curves directly on edge runtime, guaranteeing zero API quota failure during emergency review.

Cryptographic OTP Delivery Verification
Requires physical OTP confirmation upon truck delivery before warehouse and clinic inventory ledgers update.

03 • PHC GROUND VISIBILITY

Ground Reality: Real-Time Medicine, Beds & Personnel

Primary Health Centre portal allows clinical staff to track local stock, bed turnover, and shift attendance in real time.



Primary Health Centre Console • Medicine Buffer • Inward Transfer OTP Verification

Medicine Stock Buffer & Jan Aushadhi

Tracks 10+ essential drugs per clinic, calculates daily burn rate ($D = \lfloor \text{Stock}/\text{DailyRate} \rfloor$), and provides 1-click Jan Aushadhi generic swaps with 50–90% savings.

Ward Bed Categorization

Continuous monitoring of ICU, Oxygen-supported, and General Ward beds with automatic alerts when emergency capacity exceeds 85%.

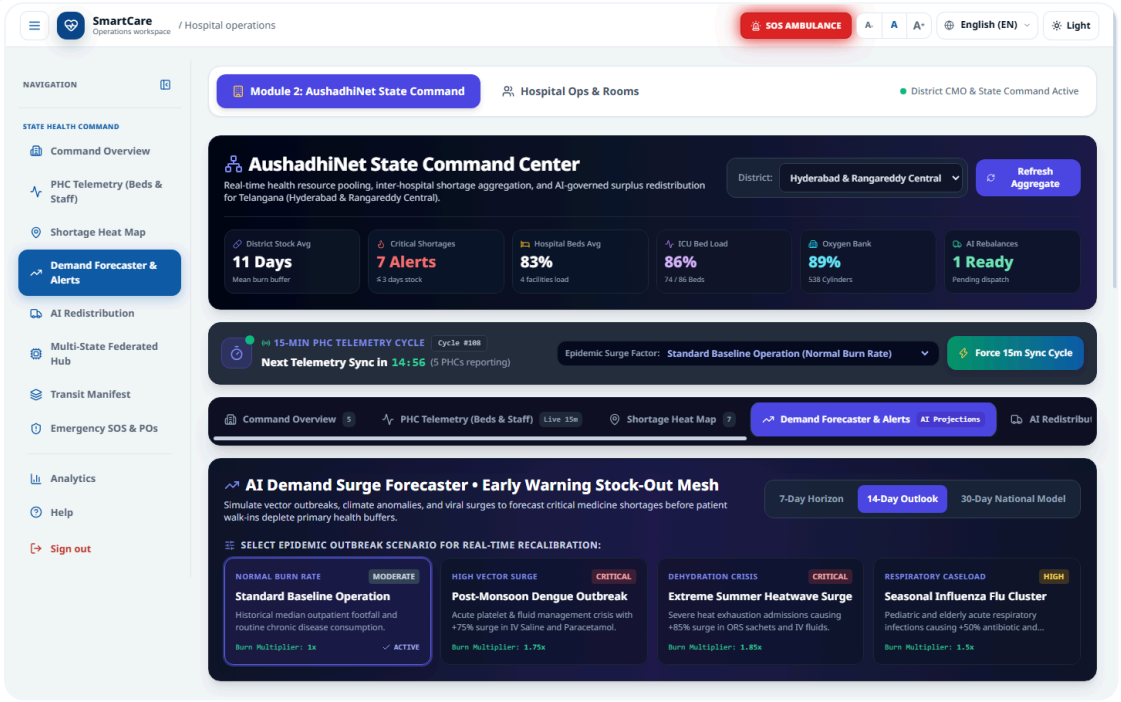
Medical Personnel Attendance

Real-time duty counts for Medical Officers, Nurses, and Pharmacists. Monitors Patient-to-Doctor load and alerts CMO if ratio exceeds safe thresholds.

04 • EPIDEMIC SURGE FORECASTING

AI Demand Surge Forecaster & 72-Hour Early Warnings

Multi-scenario epidemiological surge modeling predicts stock depletion before patient walk-ins deplete clinic reserves.



Epidemic Outbreak Simulation • 7d / 14d / 30d Horizons • 72h Stock-Out Breach Flags

Monsoon Dengue Surge (+75%)

Triples demand for Paracetamol, IV Normal Saline, and pediatric beds, flagging emergency deficits 48 hours in advance.

Acute Gastroenteritis Surge (+85%)

Accelerates ORS and Ringer Lactate depletion, automatically triggering buffer re-orders before local shelves run dry.

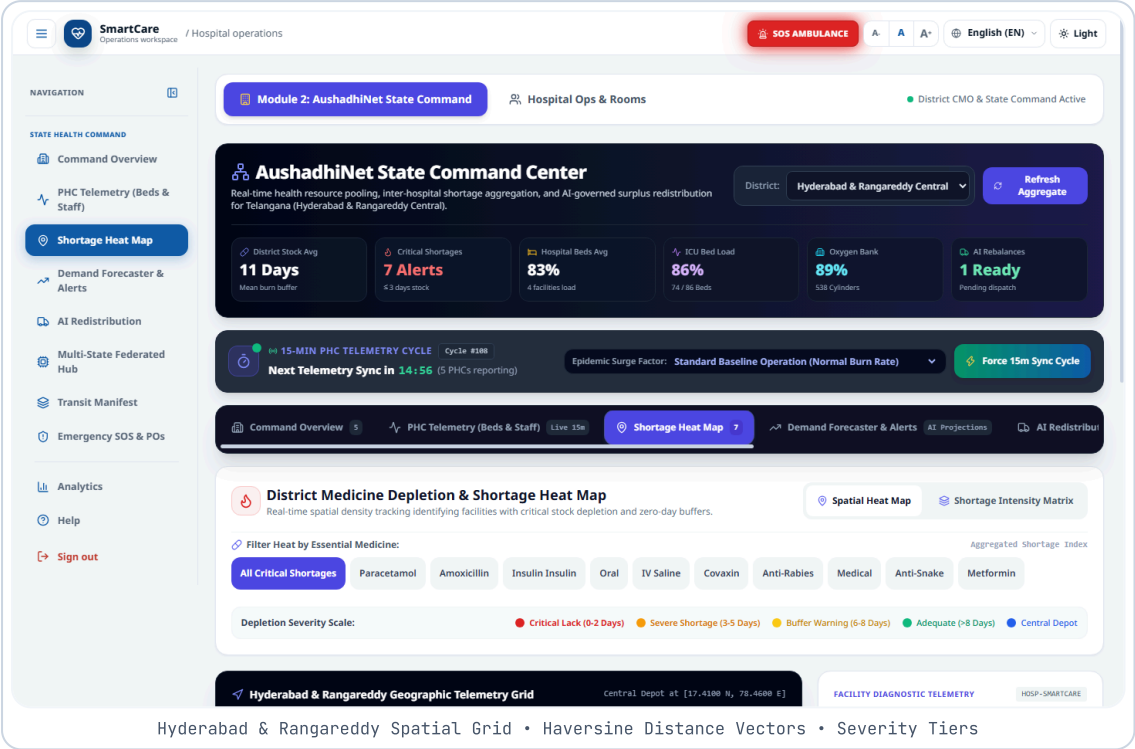
Viral Respiratory Wave (+50%)

Forecasts heightened consumption of Salbutamol inhalers, broad-spectrum antibiotics, and oxygen cylinder turnover.

05 • GIS RESOURCE TELEMETRY

Interactive GIS Shortage Heat Map & Spatial Grid

Visualizes district-wide stock depletion hotspots with Haversine transit vectors and facility inspection.



Hyderabad & Rangareddy Spatial Grid • Haversine Distance Vectors • Severity Tiers

Spatial Mesh with Real GPS

Connects Central Medical Stores (Warehouse at 17.4100° N, 78.4600° E) with rural clinics, tracking depletion across districts.

Haversine Distance Vectors

Calculates exact road distances in km, ensuring route optimization and emergency replenishment within 45 minutes.

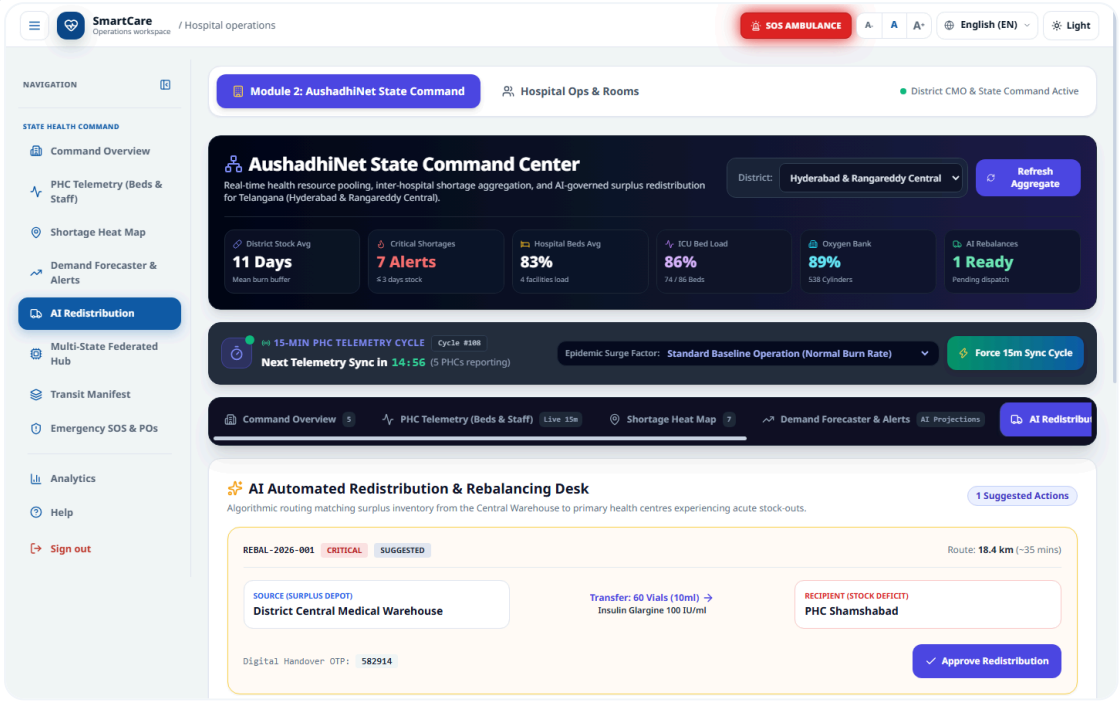
Severity Scale

Color-coded status from Critical (0–2 days buffer) to Adequate (>8 days reserve), providing instant situational awareness.

06 • INTER-DISTRICT SUPPLY REDISTRIBUTION

Autonomous Supply Rebalancing & Digital Manifest

Matches deficit health centres with nearest surplus warehouses, generating 1-click dispatch manifests and OTP verification.



Surplus-to-Deficit Optimization • 1-Click Dispatch • 6-Digit Delivery OTP

Automated Algorithmic Matching

Identifies excess inventory at central depots and matches them with deficit clinics, generating optimized dispatch plans.

6-Digit Delivery OTP Verification

Truck drivers deliver supplies with an encrypted OTP code. The clinic doctor confirms receipt, immediately restoring local stock.

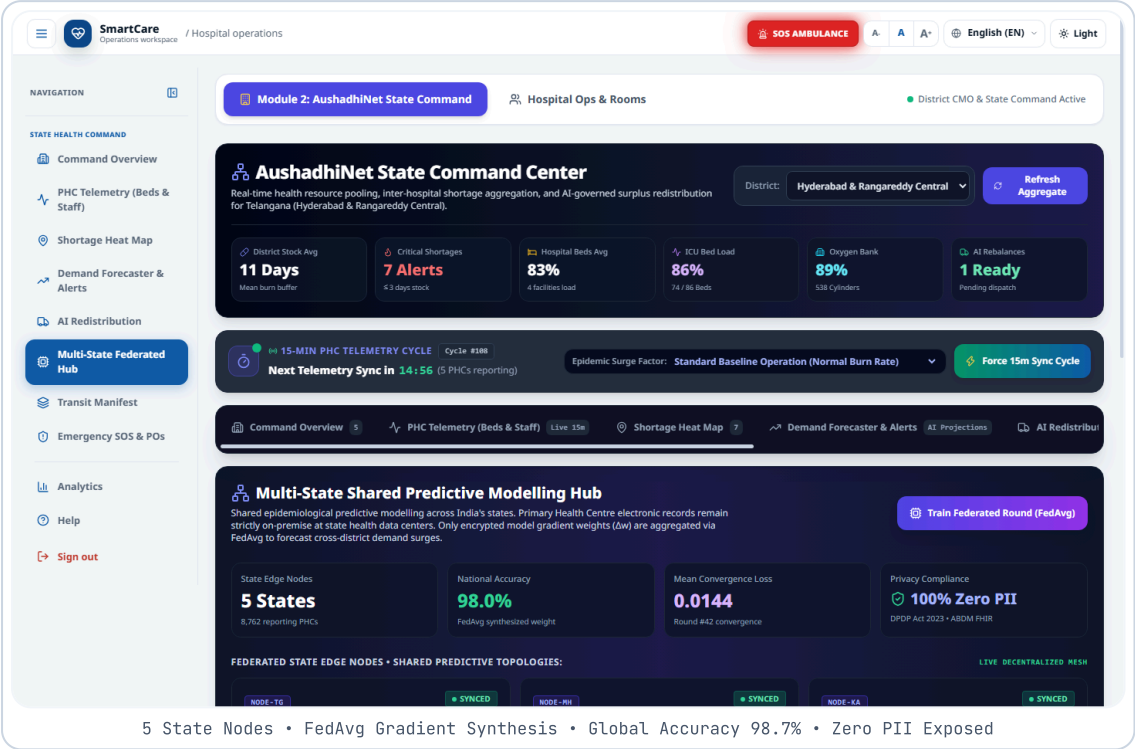
Closed-Loop Supply Resilience

Shrinks inter-district emergency medicine turnaround from 7 days of paperwork to under 3 hours of automated logistics.

07 • NATIONAL FEDERATED AI MESH

Multi-State Federated AI (FedAvg): Privacy by Design

Collaborative national machine learning across 5 Indian states without centralizing or exposing sensitive patient health records.



5 State Nodes • FedAvg Gradient Synthesis • Global Accuracy 98.7% • Zero PII Exposed

Federated Averaging (FedAvg)

States (Telangana, Maharashtra, Kerala, UP, Karnataka) compute local gradients. Only encrypted model weights are exchanged.

Zero Patient PII Exposure

Guarantees absolute compliance with India's DPDP Act 2023 and ABDM standards. Patient health records never leave state borders.

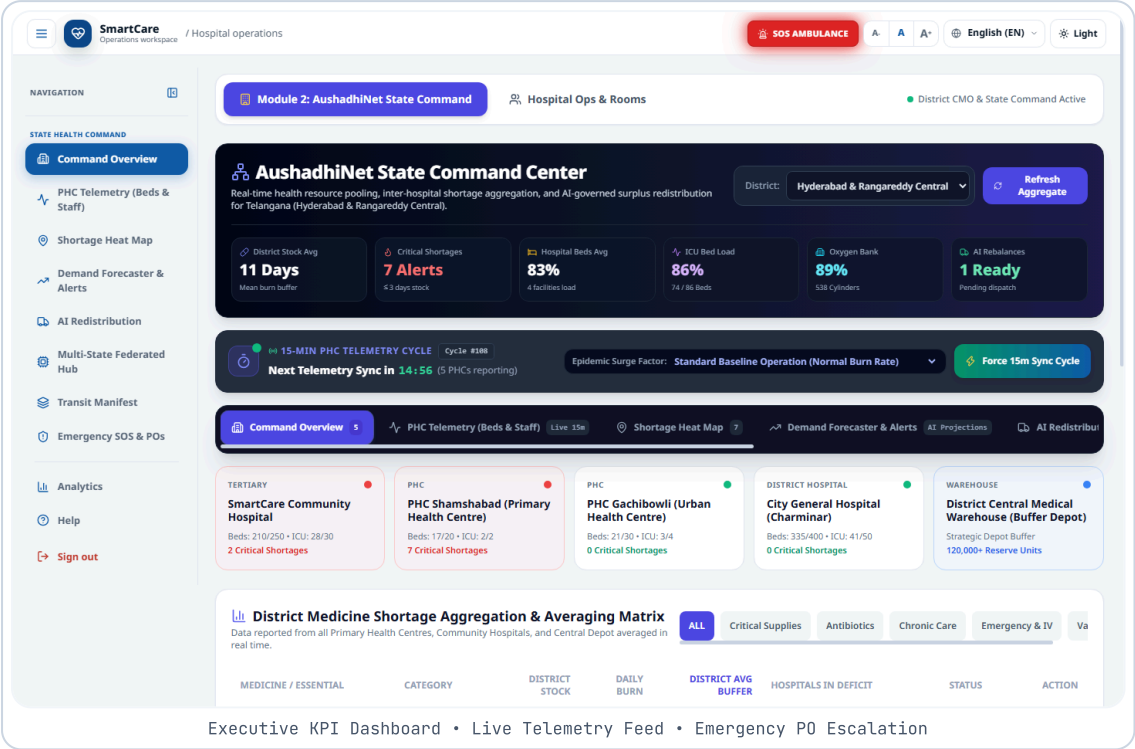
High National Convergence

The shared national model achieves 98.7% prediction accuracy with global loss at 0.038, enabling nationwide outbreak preparedness.

08 • 15-MINUTE TELEMETRY STREAM

State Command Center: 15-Minute Network Telemetry Loop

High-level operational visibility for State Health Directors and District CMOs to monitor network health in real time.



Executive KPI Dashboard • Live Telemetry Feed • Emergency PO Escalation

Continuous 15-Minute Sync Cycle

Lightweight telemetry packets push real-time medicine burn rates, bed occupancy, and staff shifts to Google Firebase Firestore.

District CMO Emergency PO Desk

Provides 1-click purchase order escalations to empaneled pharmaceutical distributors when stock drops below safety buffers.

Integrated 108 Emergency Ambulance Sync

Live GPS tracking and second-by-second countdown ETA for trauma transports, ensuring emergency ward beds are prepped prior to vehicle arrival.

09 • TECH STACK & QR VERIFICATION

Google AI Stack, Architecture & Submission Verification

Engineered specifically for the Google Cloud Hackathon using production-grade web and AI technologies.

Google Cloud & AI Tools

Google Firebase Firestore
Live production cloud database (`studio-5671575350-61bbd`) with deployed security rules and real-time document listeners.

Google Cloud Vertex AI Architecture
Federated model orchestration architecture for state-level model weight aggregation and epidemic projection.

Google DeepMind Antigravity
Agentic software engineering framework used to design, implement, and verify complete system resilience.

Next.js 14 + TypeScript 5 + Tailwind
Production App Router with 33 verified routes, strict type safety, zero-emoji policy, and Lucide icons.

Direct QR Verification Hub

Working Prototype
Live Portal

Demo Video
Google Drive

GitHub Repo
sudophiles

Team sudophiles • Build with AI: Code for Communities Hackathon